

Assignment 3: Real-World Data Engineering & Insights

Course: Programming for Artificial Intelligence

Dataset: Online Retail Dataset II

1. Objective

To master the end-to-end data pipeline: from loading "dirty" commercial data to performing statistical inference and advanced aggregation using Pandas and SciPy.

2. Dataset Setup

This is a transactional dataset which contains all the transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based and registered non-store online retail.

- **Columns:**
InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country.

3. Requirements

Task 1: Data Auditing & Cleaning (Missing Data)

Real-world data is rarely complete. You must audit the dataset and handle anomalies.

1. **Missing Values:** Identify which columns have missing values.
 - a. Drop rows where CustomerID is missing (since we cannot track loyal customers without it).
 - b. Use forward fill for rows with missing values in date and time columns
2. **Negative Values:** The Quantity column contains negative values (representing returns/cancellations). Create a separate DataFrame for "Returns" and remove them from the "Main Sales" DataFrame.
3. **Data Typing:** Convert InvoiceDate to a proper Pandas datetime object.

Task 2: Feature Engineering & Merging

1. **Total Cost:** Create a new column Total_Spent by multiplying Quantity and UnitPrice.
2. **Time Extraction:** Extract the Month and DayOfWeek from the InvoiceDate.
3. **Merging:** Use a mapping table (to be created by you) to categorize the top 10 Countries into "Regions" (e.g., UK, Europe, Rest of World) using pd.merge() or .map().

Task 3: GroupBy & Pivot Table Analytics

1. **Customer Value:** Use groupby to find the top 5 customers by Total_Spent.
2. **Sales Heatmap:** Create a pivot_table showing the average Total_Spent with DayOfWeek as rows and Month as columns.
3. **Regional Insights:** Find the average unit price of items sold per Region.

Task 4: Statistical Inference & Outliers

1. **Outlier Detection:** Use **Z-Scores** (from scipy.stats) to identify transactions with an unusually high Quantity (more than 3 standard deviations from the mean).
2. **Hypothesis Testing:** Is there a significant difference in the average Total_Spent between the UK and the rest of Europe?
 - Define H_0 .
 - Perform an **Independent T-Test**.
 - Report the **P-value** and conclude your findings at a 0.05 significance level.

Submission Requirements

1. **Source Code:** Well-commented Python script/notebook documenting each step.
2. **Report:** A brief PDF summary (1- 2 page) answering:
 - What percentage of data was lost during cleaning?
 - Which day of the week is busiest for the retailer?
 - Is the performance difference between UK and Europe statistically significant?

4. Implementation Details & Constraints

- **Forbidden:** The use of for loops to iterate through rows is strictly prohibited. You must use Vectorized Pandas operations.
- **Efficiency:** Points will be deducted for inefficient code (e.g., unnecessary copies of large DataFrames).
- **Visualization:** Include at least two plots (using matplotlib or seaborn) to support your Task 3 and Task 4 findings.